

Abstract—High-speed moving target detection is an important task for airborne radar. However, it inevitably encounters the problem of Doppler ambiguity and interference from ground clutter. Existing methods for resolving Doppler ambiguity and suppressing clutter are relatively mature, however approaches that simultaneously address Doppler ambiguity resolution and clutter suppression are relatively scarce. This paper proposes a strategy that uses frequency diverse array multiple-input multiple-output space-time adaptive processing (FDA-MIMO-STAP) radar to address the Doppler ambiguity problem in high-speed moving target detection. By introducing the Doppler spreading property of FDA-MIMO into STAP, Doppler-ambiguous clutter is successfully suppressed. Meanwhile, we also derive a lower bound on the Doppler-ambiguous clutter suppression performance of the FDA-MIMO radar. In addition, by utilizing the proposed strategy, we accomplish the task of Doppler ambiguity resolution for high-speed targets while suppressing Doppler-ambiguous clutter. Experimental results validate the correctness of the derived bound on Doppler-ambiguous clutter suppression performance for the FDA-MIMO radar and the effectiveness of the proposed strategy for resolving Doppler ambiguity of high-speed targets in airborne clutter environments.

Index Terms—High-speed moving target, Doppler ambiguity, frequency diverse array multi-input and multi-output radar, Doppler spreading, Doppler ambiguous clutter suppression

I. INTRODUCTION

THE airborne early warning (AEW) radar is a critical type of detection radar in modern warfare. Owing to its wide detection range and flexible detection modes, it significantly extends the warning scope of fighter aircraft and has garnered considerable attention from researchers [1,2]. When the airborne radar operates in a downward-looking mode, the received target echoes are inevitably mixed with ground and sea clutter, which is attributed to the wide and focused downward beam of radar. Due to the high-speed movement of the airborne platform, the received clutter also exhibits an additional Doppler frequency shift. Conventional clutter suppression techniques, such as the moving target indicator (MTI), are ineffective in eliminating such clutter. Consequently, the space-time adaptive processing (STAP) approach was developed. Through two-dimensional joint processing of the spatial and Doppler domains, STAP can effectively mitigate the clutter received by the airborne platform [3]–[7]. However, when the airborne radar detects high-speed targets, the pulse Doppler (PD) radar inherently struggles to resolve Doppler ambiguity and range ambiguity simultaneously [8,9]. As a result, in addition to the

real clutter, ambiguous clutter can also enter the radar mainlobe. Under such circumstances, the airborne radar can hardly suppress the ambiguous clutter relying solely on STAP.

A common strategy to counteract ambiguous clutter is to increase the pulse repetition frequency (PRF), with most research efforts focusing on clutter suppression under medium-PRF and high-PRF conditions. Chen et al. proposed a two-stage adaptive processing method based on stepped-frequency radar, which separates range-ambiguous clutter of different orders by partitioning the clutter in the range-Doppler domain [10]. Wang et al. introduced a method that combines an improved Doppler warping (IDW) technique with clutter covariance matrix (CCM) reconstruction [11]. It reconstructs the clutter data and CCMs for all ambiguous range rings based on the radar-target geometry, but is limited to bistatic radar configurations. He et al. presented a framework based on an elevation time-diverse array (TDA) and azimuth phase coding (APC) to mitigate range-ambiguous clutter [12], which exploits a joint TDA-APC waveform design to discriminate clutter from different range ambiguities. Qiu et al. proposed a phase-coding method in the pulse dimension [13]. It exploits the differences in coherent integration gain of signals across different range ambiguity intervals to separate ambiguous clutter. Furthermore, Wang et al. designed an element-pulse jointly coded MIMO radar [14]. It relies on the favorable orthogonality of extended phase codes (EPC) to separate clutter from different ambiguous range regions in the transmit-Doppler domain. Duan et al. employed a scheme combining slow-time frequency coding with phase-modulated waveforms [15]. It utilizes the unique joint time-frequency signatures arising from pulse-to-pulse variations to distinguish signals from different ambiguous range intervals. The performance of these waveform-coding-based methods hinges critically on waveform orthogonality, suffering substantial degradation when orthogonality is compromised. To address this sensitivity, Lu et al. proposed a method that integrates random phase coding with subspace projection filtering, suppressing ambiguous clutter with random phase coding alone [16]. However, its practical application is restricted by the requirement for prior knowledge of the clutter range ambiguity order.

Frequency diverse array (FDA) radar acquires range-dimensional degrees of freedom by introducing a small frequency increment between adjacent array elements [17] and exhibits distinct advantages in target detection [18,19], parameter estimation [20,21] and interference suppression [22,23]. Benefiting from such range-dimensional degrees of freedom, FDA can also be applied to suppress range-ambiguous clutter. Wang et al. proposed a FDA transmit mainlobe correction and receive inverse-compensation method. By exploiting the range-coupled phase differences of clutter with different ambiguity number after correction and compensation, clutter corresponding to different range ambiguity number can be separated [24]. Song et al. presented a subspace-based FDA-

STAP method. By estimating the clutter rank under range ambiguity, a subspace projection filter for ambiguous clutter is constructed to achieve the suppression of range-ambiguous clutter [25]. Sun et al. approached the problem from the perspective of second-order range ambiguity in FDA and developed an optimized nonlinear frequency-increment scheme to suppresses range-ambiguous clutter [26]. By introducing frequency diverse array multi-input and multi-output (FDA-MIMO) radar, Wu et al. constructed a three-dimensional overcomplete dictionary in the range-angle-Doppler domain and realized range-ambiguous clutter suppression by adopting compressed sensing technology [27]. Existing FDA-based methods for range-ambiguous clutter suppression are generally applicable only to scenarios with a small number of ambiguity orders. Accordingly, Qiu et al. proposed an improved element-pulse coded FDA-MIMO radar framework combined with sliding subarray time-division transmission technology, which is capable of dealing with ambiguous clutter under extremely high range ambiguity orders [28]. Nevertheless, the ambiguous clutter suppression performance of this method is constrained by waveform orthogonality.

Inspired by the blind Doppler target detection method of Gui et al. [29], we propose a method for suppressing ambiguous clutter based on Doppler spreading of FDA-MIMO radar. Different from the research on range ambiguous clutter suppression in high PRF scenarios, we convert the range ambiguous clutter into Doppler-ambiguous clutter by reducing the PRF. Leveraging the Doppler spreading capability of FDA-MIMO radar, the proposed method could suppresses Doppler-ambiguous clutter while simultaneously resolving Doppler ambiguity for high-speed targets. First, we establish the target echo and clutter models in the airborne FDA-MIMO radar down-looking operation and high-speed target detection scenarios. Then, we analyze the clutter signal with Doppler spreading in detail and derive the lower bound of the Doppler-ambiguous clutter suppression performance of FDA-MIMO-STAP radar. Further, we also analyze the lower bound of the Doppler-ambiguity period of FDA-MIMO radar and prove that FDA-MIMO-STAP radar could both suppress Doppler-ambiguous clutter and resolve the Doppler ambiguity of high-speed targets. Our contributions are as follows:

- Based on the complete process of signal transmission, propagation and reception, we derive the target echo and clutter signal models considering the Doppler spreading when the airborne FDA-MIMO radar is operating in the down-looking mode.
- Based on the analysis of FDA-MIMO clutter signals, we derive the closed-form relationship between the first-order sidelobe and frequency increment of the FDA-MIMO-STAP radar clutter spectrum and determine the theoretical lower bound of Doppler-ambiguous clutter suppression for FDA-MIMO-STAP radar under specific frequency incre-

ment. The experiments verify the correctness of the derived results.

- Based on the analysis of the FDA-MIMO target echo signal model, we prove that the FDA-MIMO radar has the ability to resolve the Doppler ambiguity of high-speed targets and derive the theoretical lower bound of its Doppler ambiguity resolution.

The rest of this paper is organized as follows. Section II provides the target and clutter signal models of FDA-MIMO radar. In Section III, we analyze the performance of Doppler-ambiguous clutter suppression using FDA-MIMO-STAP radar. In Section IV, we propose a strategy of resolving Doppler ambiguity for high-speed moving target detection and derive the corresponding lower bound of resolving Doppler ambiguity. The experiment results in Section V demonstrate the validity of our formulated FDA-MIMO clutter ambiguity sidelobe. Finally, the paper is concluded in Section VI.

Notations: Boldface is used for vectors $\mathbf{x}$ (lower case), and matrices $\mathbf{A}$ (upper case). Superscripts $(\cdot)^T$, $(\cdot)^H$ and $(\cdot)^{-1}$ denote the operation of transpose, conjugate transpose and matrix inverse, respectively. The Kronecker product and the Hadamard product are indicated as $\otimes$ and $\odot$, respectively. The notation $\sim$ means "be distributed as". $\text{diag}\{\cdot\}$ output a diagonal matrix formed by the elements inside the braces. $\text{vec}\{\cdot\}$ represents the operation of vectorization. $\mathbb{C}^{m \times n}$ is the $(m \times n)$ -dimensional complex matrix and $\mathbb{Z}^+$ denotes the set of positive integers. $\mathcal{CN}(\mu, \sigma^2)$ denotes the circularly symmetric, complex-valued Gaussian distribution with mean μ and covariance matrix σ^2 . $\mathbf{1}$ is the all-ones matrix, $\mathbf{0}$ is the all-zeros matrix and $\mathbf{I}$ is the identity matrix.

II. FDA-MIMO SIGNAL MODEL

In this section, the airborne FDA-MIMO radar signal model will be formulated. As shown in Fig.1, there is a geometric scenario of the ground observation of an airborne radar. A Cartesian coordinate system is established with the center of the radar as the origin. Assume that the operating altitude and speed of the aircraft are H_a and v_a , respectively. And the direction of radar movement is consistent with the y-axis direction. The target and clutter scattering patches are far away from the radar. The airborne radar is considered to be a FDA-MIMO array with side-looking configuration. The transmitting antenna and receiving antenna are co-located, with the number of antennas being M and N , respectively. The inter-spacing of FDA-MIMO radar is presented as d and it satisfies $d = \lambda_0/2$ where λ_0 presents the wavelength of radar signal. Correspondingly, the transmitted signal of m -th transmit element is presented as

$$s_m(t) = s(t)e^{j2\pi f_m t}, m = 0, \dots, M-1 \quad (1)$$

where $f_m = f_0 + m\Delta f$ is the carrier frequency of the m -th element and $s(t)$ is the baseband waveform with great autocorrelation characteristics. Meanwhile, Δf is

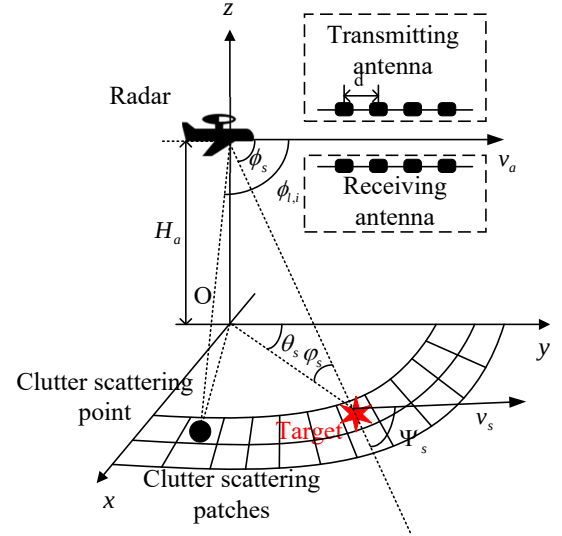


Fig. 1: The geometric scene of airborne radar operation

greater than the bandwidth of $s(t)$ which meets the MIMO condition.

A. FDA-MIMO target signal model

Suppose that there is a moving target in the observation region of radar. And its position relative to the radar reference coordinate can be expressed as (ϕ_s, r_s) , where ϕ_s represents the line-of-sight (LOS) angle, and r_s represents the slant range. Meanwhile, the target is moving at a speed of v_s and forms an angle of Ψ_s with the direction of radar LOS. According to the geometric relationship shown in Fig.1, the azimuth angle θ_s and elevation angle φ_s of the target relative to the radar satisfy $\cos(\phi_s) = \cos(\theta_s)\cos(\varphi_s)$. Assume that the signal is transmitted by the m -th element, back-scattered by the target, and received by the n -th element. The propagation delay can be expressed as

$$r_{m,n,k} = 2r_s - 2(k-1)T_r(v_a \cos(\phi) - v_s \cos(\Psi_s)) - (m-1)d \cos(\phi_s) - (n-1)d \cos(\phi_s) \quad (2)$$

In (2), $\{k\}_1^K$ presents the index of pulses and K is the number of pulses in a coherent processing interval (CPI). T_r stands for the pulse repetition time (PRT). And then, the received signal of n -th receiving element k -th pulse is expressed as

$$y_{n,k}(t; r_s, \phi_s) = \xi_s \sum_{m=1}^M s_m \left(t - \frac{r_{m,n,k}}{c} \right) \quad (3)$$

ξ_s denotes the back-scattering coefficient of the target. c is the speed of light. Through multi-channel matched filtering[gui], $y_{n,k}(t; r_s, \phi_s)$ is separated into M channel signals. The signal in n -th receiving element m -th channel

is indicated as

$$\begin{aligned}
y_{m,n,k}(t; r_s, \phi_s) &= \int_{t_0}^{t_0+T_{obs}} y_{n,k}(\tau) \mu_s s_m(t-\tau) e^{-j2\pi f_0 t} d\tau \\
&\approx \xi_s e^{j2\pi(m-1)\left(\frac{dT}{\lambda_0} \cos(\phi_s) - \frac{2r_s}{c} \Delta f\right)} \\
&\quad \times e^{j4\pi \frac{(k-1)(v_a \cos(\phi) - v_s \cos(\Psi_s))T_r}{c} f_m} \\
&\quad \times e^{j2\pi \frac{(n-1)d_R}{\lambda_0} \cos(\phi_s)} \gamma_s\left(t - \frac{r_s}{c}\right)
\end{aligned} \quad (4)$$

where t_0 , T_{obs} represent the front edge and the length of the observation window, respectively. μ_s denotes the amplitude coefficient of matching filter. The above approximate relationship holds under the condition of narrow bandwidth. And $\gamma_s(t)$ is expressed as

$$\gamma_s(t - t_r) \triangleq \mu_s \int_{-\infty}^{\infty} s(\tau - t) s(\tau - t_r) d\tau \quad (5a)$$

$$\int_{-\infty}^{\infty} |\gamma_s(t)|^2 dt = 1 \quad (5b)$$

By sampling $y_{m,n,k}(t; r_s, \phi_s)$ with sample frequency f_s , $y_{m,n,k}(t; r_s, \phi_s)$ can be rewritten as follows

$$\begin{aligned}
y_{m,n,k}[l; l_{r,s}, \phi] &= y_{m,n,k}(l/f_s + t_0; r_s, \phi_s), l = \{1, \dots, L\} \\
l_{r,s} &= [(r_s/c - t_0)f_s]
\end{aligned} \quad (6)$$

where L denotes the number of sampling points and it satisfies $L = [T_{obs}f_s]$. Due to the great autocorrelation property of $s(t)$, the energy of $y_{m,n,k}[l; l_{r,s}, \phi_s]$ is concentrated at $l = l_{r,s}$.

Furthermore, combining $y_{m,n,k}[l; l_{r,s}, \phi_s]$ of multiple channels, we can obtain vectorized two-dimensional space time sampling data as follows:

$$\begin{aligned}
\mathbf{y}_s &= [y_{1,1,1}[l; l_{r,s}, \phi_s], \dots, y_{m,n,k}[l; l_{r,s}, \phi_s], \\
&\quad \dots, y_{M,N,K}[l; l_{r,s}, \phi_s]]^T
\end{aligned} \quad (7)$$

To simplifying derivation, we define transmitting spatial frequency f_T , receiving spatial frequency f_R and Doppler frequency f_D as follows:

$$f_T(\phi, r) \triangleq \frac{dT}{\lambda_0} \cos(\phi) - \frac{2r}{c} \Delta f \quad (8a)$$

$$f_R(\phi) \triangleq \frac{dR}{\lambda_0} \cos(\phi) \quad (8b)$$

$$f_{D,m}(v) \triangleq \frac{vT_r}{c} f_m \quad (8c)$$

Then, (7) can be rewritten as

$$\begin{aligned}
\mathbf{y}_s &= \xi_s \mathbf{vec}\{\mathbf{A}_{TRD}(\phi_s, r_s, v_a \cos(\phi_s) \\
&\quad - v_s \cos(\Psi_s))\} \gamma_s[l; r_s]
\end{aligned} \quad (9)$$

where

$$\gamma_s[l; r] = \gamma_s\left(\frac{l}{f_s} + t_0 - \frac{r}{c}\right), l = \{1, \dots, L\} \quad (10a)$$

$$\begin{aligned}
\mathbf{A}_{TRD}(\phi, r, v) &\triangleq \\
&\quad [\text{diag}\{\mathbf{a}_T(f_T(\phi, r))\} \otimes \mathbf{a}_R(f_R(\phi))] \mathbf{A}_D(v)
\end{aligned} \quad (10b)$$

$$\mathbf{a}_T(f_T(\phi, r)) \triangleq [1, e^{j2\pi f_T}, \dots, e^{j2\pi(M-1)f_T}]^T \quad (10c)$$

$$\mathbf{a}_R(f_R(\phi)) \triangleq [1, e^{j2\pi f_R}, \dots, e^{j2\pi(N-1)f_R}]^T \quad (10d)$$

$$\begin{aligned}
\mathbf{A}_D(v) &\triangleq [\mathbf{a}_D(f_{D,1}(v)), \\
&\quad \mathbf{a}_D(f_{D,2}(v)), \dots, \mathbf{a}_D(f_{D,M}(v))]^T
\end{aligned} \quad (10e)$$

$$\mathbf{a}_D(f_{D,m}(v)) \triangleq [1, e^{j2\pi f_{D,m}}, \dots, e^{j2\pi(K-1)f_{D,m}}]^T \quad (10f)$$

The vectorized sampling data $\mathbf{y}_s$ satisfies $\mathbf{y}_s \in \mathbb{C}^{MNK \times L}$. $\{\mathbf{a}_T \in \mathbb{C}^M, \mathbf{a}_R \in \mathbb{C}^N, \mathbf{A}_D \in \mathbb{C}^{M \times K}, \mathbf{a}_D \in \mathbb{C}^K\}$ represent transmitting steering vector, receiving steering vector, Doppler spreading matrix and Doppler vector in m -th channel, respectively.

B. FDA-MIMO clutter signal model

It is assumed that there are N_c clutter scattering points in l -th range patch. Each clutter scattering point is equivalent to a target with $v = 0$. Correspondingly, the LOS angle of the i -th clutter cell in l -th range patch relative to the radar is $\phi_{l,i}$ and the echo amplitude is $\xi_{l,i} \sim \mathbb{N}(0, \sigma_c^2)$. Then, the clutter within the i -th range patch can be expressed as:

$$\mathbf{y}_c = \sum_{l=1}^L \sum_{i=1}^{N_c} \xi_{l,i} \mathbf{vec}\{\mathbf{A}_{TRD}(\phi_{l,i}, r_l, v_a \cos(\phi_{l,i}))\} \gamma_s[l; r_l] \quad (11)$$

For FDA-MIMO radar, due to the effect of range-dependent characteristic, the distribution of clutter scattering points in each range patch is not uniform. Therefore, it is necessary to construct a compensation vector in the transmitting spacial domain to compensate it for subsequent processing. The compensation vector $\mathbf{h}(r_l)$ can be expressed as

$$\mathbf{h}(r_l) = [1, e^{j4\pi \frac{r_l}{c} \Delta f}, \dots, e^{j4\pi(M-1) \frac{r_l}{c} \Delta f}]^T \otimes \mathbf{1}_{NK} \quad (12)$$

By compensating for the clutter data $\mathbf{y}_c$, the clutter data $\bar{\mathbf{y}}_c$ without range-dependence can be expressed as

$$\begin{aligned}
\bar{\mathbf{y}}_c &= \sum_{l=1}^L \sum_{i=1}^{N_c} \xi_{l,i} [\mathbf{h}(r_l) \odot \mathbf{vec}\{\mathbf{A}_{TRD}(\phi_{l,i}, r_l, \\
&\quad v_a \cos(\phi_{l,i}))\}] \gamma_s[l; r_l] \\
&= \sum_{l=1}^L \sum_{i=1}^{N_c} \xi_{l,i} \mathbf{vec}\{\mathbf{A}_{TRD}(\phi_{l,i}, 0, v_a \cos(\phi_{l,i}))\} \gamma_s[l; r_l]
\end{aligned} \quad (13)$$

Considering that clutter scattering points are far away from the radar, the propagation delay $2r_{l,i}/c$ is much longer than the length of observation window T_{obs} . The elevation angles of L clutter scattering points with the same azimuth angle θ_i satisfy the relationship as follows

$$\begin{aligned}\varphi_{l,i} &= \arcsin\left(\frac{H_a}{r_{l,i}}\right) \approx \arcsin\left(\frac{H_a}{r_{1,i}}\right) \\ &= \varphi_{1,i} = \varphi_i\end{aligned}\quad (14)$$

In that case, the LOS angle $\phi_{l,i}$ satisfies the following relationship.

$$\begin{aligned}\phi_{l,i} &= \arccos(\cos(\theta_i)\cos(\varphi_{l,i})) \\ &\approx \arccos(\cos(\theta_i)\cos(\varphi_i)) = \phi_i\end{aligned}\quad (15)$$

Substituting (15) into (13), (13) can be rewritten as

$$\begin{aligned}\bar{\mathbf{y}}_c &= \sum_{i=1}^{N_c} \text{vec}\{\mathbf{A}_{TRD}(\phi_i, 0, v_a \cos(\phi_i))\} \sum_{l=1}^L \xi_{l,i} \gamma_s[l; r_l] \\ &\approx \sum_{i=1}^{N_c} \text{vec}\{\mathbf{A}_{TRD}(\phi_i, 0, v_a \cos(\phi_i))\} \boldsymbol{\xi}_i^H\end{aligned}\quad (16)$$

where,

$$\boldsymbol{\xi}_i = [\xi_{1,i}, \dots, \xi_{L,i}, \dots, \xi_{L,i}]^H \in \mathbb{C}^L, \boldsymbol{\xi}_i \sim \mathcal{CN}(0, \sigma_c^2)\quad (17)$$

The approximation in (16) is established based on the strong autocorrelation property of $s(t)$ and the normalized energy relationship of (5b). For the subsequent processing, $\bar{\mathbf{y}}_c$ can be expressed in the form of matrix multiplication as follows

$$\bar{\mathbf{y}}_c = \mathbf{S}_c \mathbf{X}_c\quad (18)$$

where,

$$\mathbf{S}_c = [\mathbf{a}_{trd}(\phi_1), \dots, \mathbf{a}_{trd}(\phi_i), \dots, \mathbf{a}_{trd}(\phi_{N_c})]\quad (19a)$$

$$\mathbf{a}_{trd}(\phi_i) = \text{vec}\{\mathbf{A}_{TRD}(\phi_i, 0, v_a \cos(\phi_i))\}\quad (19b)$$

$$\mathbf{X}_c = [\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_i, \dots, \boldsymbol{\xi}_{N_c}]^T\quad (19c)$$

III. FDA-MIMO AMBIGUITY CLUTTER SUPPRESSION

In this section, to better analyze characteristics of FDA-MIMO clutter, only the clutter scattering patches are considered in the scenario of Fig. 1. Correspondingly, the received signal by FDA-MIMO radar can be expressed as

$$\begin{aligned}\mathbf{y} &= \bar{\mathbf{y}}_c + \mathbf{y}_n, \\ \mathbf{y}_n &\in \mathbb{C}^{MNK \times L}, \mathbf{y}_n \sim \mathcal{CN}(0, \sigma_n^2)\end{aligned}\quad (20)$$

where $\mathbf{y}_n$ denotes the additive Gaussian white noise inside the receiver. Furthermore, the clutter covariance matrix can be calculated.

$$\begin{aligned}\mathbf{R}_c &= \frac{1}{L} \bar{\mathbf{y}} \bar{\mathbf{y}}^H \\ &= \frac{1}{L} \mathbf{S}_c \mathbf{X}_c \mathbf{X}_c^H \mathbf{S}_c^H + \sigma_n^2 \mathbf{I}_{MNK} \\ &= \mathbf{S}_c \mathbf{C} \mathbf{S}_c^H + \sigma_n^2 \mathbf{I}_{MNK}\end{aligned}\quad (21)$$

where,

$$\mathbf{C} = \begin{bmatrix} \rho_{1,1} & \cdots & \rho_{1,N_c} \\ \vdots & \ddots & \vdots \\ \rho_{N_c,1} & \cdots & \rho_{N_c,N_c} \end{bmatrix} \in \mathbb{C}^{N_c \times N_c}\quad (22a)$$

$$\rho_{i,j} = \frac{1}{L} \boldsymbol{\xi}_i^H \boldsymbol{\xi}_j \approx \begin{cases} \sigma_c^2, i=j \\ 0, i \neq j \end{cases}\quad (22b)$$

When the clutter scattering points with different ϕ are independent of each other, the approximate condition in (22b) is satisfied. In that case, $\mathbf{C}$ degenerates into a diagonal matrix. Therefore, the clutter with different LOS angle ϕ exhibits a weak mutual influence. To simplifying the analysis, we only resolve the effect of scattering patches with the same LOS angle ϕ_i which can characterize the overall clutter. Substituting ϕ_i into (21), $\mathbf{R}_c$ can be rewritten as

$$\mathbf{R}_{c,i} = \sigma_c^2 \mathbf{a}_{trd}(\phi_i) \mathbf{a}_{trd}^H(\phi_i) + \sigma_n^2 \mathbf{I}_{MNK}\quad (23)$$

To analyze the effect of FDA-MIMO Doppler spreading on clutter characteristics, we need to obtain the space-time distribution of the clutter. The conventional approach is to use time-space adaptive processing (STAP) to solve for the optimal weight vector in a specific LOS angle ϕ and velocity v as follows:

$$\begin{cases} \min \boldsymbol{\omega}^H \mathbf{R}_{c,i} \boldsymbol{\omega} \\ \text{s.t. } \boldsymbol{\omega}^H \boldsymbol{\beta}(\phi, v) = 1 \end{cases}\quad (24)$$

where,

$$\boldsymbol{\beta}(\phi, v) \triangleq \text{vec}\{\mathbf{A}_{TRD}(\phi, 0, v)\}\quad (25)$$

By solving the above optimization problem, the optimal weight vector can be obtained as

$$\boldsymbol{\omega}_{opt} = \frac{\mathbf{R}_{c,i}^{-1} \boldsymbol{\beta}(\phi, v)}{\boldsymbol{\beta}(\phi, v)^H \mathbf{R}_{c,i}^{-1} \boldsymbol{\beta}(\phi, v)}\quad (26)$$

Furthermore, the power of (20) in the specific LOS angle ϕ and velocity v is expressed as follows

$$P(\phi, v) = \frac{1}{|\boldsymbol{\beta}^H(\phi, v) \mathbf{R}_{c,i}^{-1} \boldsymbol{\beta}(\phi, v)|}\quad (27)$$

The inverse of $\mathbf{R}_{c,i}$ can be calculated according to Sherman-Morrison formula.

$$\begin{aligned}\mathbf{R}_{c,i}^{-1} &= \frac{1}{\sigma_n^2} \left[\mathbf{I}_{MNK} - \frac{\sigma_c^2}{\sigma_n^2} \mathbf{S} \left[1 + \frac{\sigma_c^2 \mathbf{S}^H \mathbf{S}}{\sigma_n^2} \right]^{-1} \mathbf{S}^H \right] \\ &= \frac{1}{\sigma_n^2} \left[\mathbf{I}_{MNK} - \frac{\sigma_c^2}{\sigma_n^2 + MNK \sigma_c^2} \mathbf{S} \mathbf{S}^H \right]\end{aligned}\quad (28)$$

where,

$$\mathbf{S} = \mathbf{a}_{trd}(\phi_i)$$

Substitute (28) into (27)

$$P(\phi, v) = \frac{\sigma_n^2 (\sigma_n^2 + MNK \sigma_c^2)}{MNK \sigma_n^2 + M^2 K^2 N^2 \sigma_c^2 - \sigma_c^2 \text{sinc}^2(\phi, v; \phi_{c,i})}\quad (29)$$

where $\Delta\phi, \Delta v$ satisfy

$$\Delta\phi = \cos(\phi_{c,i}) - \cos(\phi)\quad (30a)$$

$$\Delta v = v_a \cos(\phi_{c,i}) - v\quad (30b)$$

$$\begin{aligned}
\text{sinc3}(\phi, v; \phi_{c,i}) &= \beta(\phi, v)^H \mathbf{S} \mathbf{S}^H \beta(\phi, v) \\
&= \left(\sum_{n=0}^{N-1} e^{j2\pi n(f_R(\phi_{c,i}) - f_R(\phi))} \right)^2 \left(\sum_{m=0}^{M-1} \sum_{k=0}^{K-1} e^{j2\pi m(f_T(\phi_{c,i}, l, 0) - f_T(\phi, 0))} e^{j2\pi k(f_{D,m}(\phi_{c,i}, v_a \cos(\phi_{c,i})) - f_{D,m}(\phi, v))} \right)^2 \\
&= \left(\frac{\sin(N\pi \frac{d_R}{\lambda_0} \Delta\phi)}{\sin(\pi \frac{d_R}{\lambda_0} \Delta\phi)} \right)^2 \left(\sum_{k=0}^{K-1} e^{j2\pi \frac{2\Delta v}{\lambda_0} k T_r} \sum_{m=0}^{M-1} e^{j2\pi m \frac{2\Delta v}{c} \Delta f k T_r} e^{j2\pi \frac{m d_T \Delta\phi}{\lambda_0}} \right)^2 \\
&= \left(\frac{\sin(N\pi \frac{d_R}{\lambda_0} \Delta\phi)}{\sin(\pi \frac{d_R}{\lambda_0} \Delta\phi)} \right)^2 \left(\sum_{k=0}^{K-1} e^{j2\pi \frac{2\Delta v}{c} (f_0 + \frac{M-1}{2} \Delta f) k T_r} \frac{\sin(N\pi (k \frac{2\Delta v}{c} \Delta f T_r + \frac{d_T \Delta\phi}{\lambda_0}))}{\sin(\pi (k \frac{2\Delta v}{c} \Delta f T_r + \frac{d_T \Delta\phi}{\lambda_0}))} \right)^2
\end{aligned} \tag{31}$$

In this section, the main attention is concentrating on the suppression of Doppler ambiguity clutter. Therefore, we assume $\Delta\phi = 0$ which means $\phi = \phi_{c,i}$. Correspondingly, $\text{sinc3}(\phi, v; \phi_{c,i})$ can be rewritten as follows:

$$\begin{aligned}
\text{sinc3}(\phi_{c,i}, v; \phi_{c,i}) &= \\
N^2 \left(\sum_{k=0}^{K-1} e^{j2\pi \Phi_1(\Delta f) k \Delta v} \frac{\sin(M\pi \Phi_2(\Delta f) k \Delta v)}{\sin(\pi \Phi_2(\Delta f) k \Delta v)} \right)^2
\end{aligned} \tag{32}$$

where

$$\Phi_1(\Delta f) = \frac{2T_r}{c} (f_0 + \frac{M-1}{2} \Delta f) \tag{33a}$$

$$\Phi_2(\Delta f) = \frac{2T_r}{c} \Delta f \tag{33b}$$

By swapping the order of additive, (32) can also be rewritten as follows:

$$\begin{aligned}
\text{sinc3}(\phi_{c,i}, v; \phi_{c,i}) &= \\
N^2 \left(\sum_{m=0}^{M-1} e^{j2\pi \Phi_3(\Delta f) m \Delta v} \text{sinc2}(f_m, \Delta v) \right)^2
\end{aligned} \tag{34}$$

where,

$$\text{sinc2}(f_m, \Delta v) = \frac{\sin(K\pi \Phi_4(f_m) \Delta v)}{\sin(\pi \Phi_4(f_m) \Delta v)} \tag{35a}$$

$$\Phi_3(\Delta f) = \frac{(K-1)\Delta f T_r}{c} \tag{35b}$$

$$\Phi_4(f_m) = \frac{2f_m T_r}{c} \tag{35c}$$

In (35a), the period of $\text{sinc2}(f_m, \Delta v)$ in velocity dimension can be expressed as

$$T_{v, \text{sinc2}}(f_m) = \frac{c}{2f_m T_r}, m = \{0, \dots, M-1\} \tag{36}$$

Meanwhile, we can establish the position of mainlobe of $\text{sinc3}(\phi_{c,i}, v; \phi_{c,i})$ through the maximum pint.

$$\begin{aligned}
\text{sinc3}(\phi_{c,i}, 0; \phi_{c,i}) &= M^2 N^2 K^2 \\
&\geq \text{sinc3}(\phi_{c,i}, v; \phi_{c,i}), \forall v
\end{aligned} \tag{37}$$

According to (37), it can be determined that $\Delta v = 0$ is the mainlobe position of $\text{sinc3}(\phi_{c,i}, v; \phi_{c,i})$. Furthermore, the first-order sidelobe position of $\text{sinc3}(\phi_{c,i}, v; \phi_{c,i})$ can

be obtained from sidelobe region $\Theta_1(\Delta v)$ with the same method.

$$\begin{aligned}
&\text{sinc3}(\phi_{c,i}, v_a \cos(\phi_{c,i}) - \Delta v_{1\text{-side}}; \phi_{c,i}) \geq \\
&\text{sinc3}(\phi_{c,i}, v_a \cos(\phi_{c,i}) - \Delta v; \phi_{c,i}), \forall \Delta v \in \Theta_1(\Delta v)
\end{aligned} \tag{38}$$

where,

$$\Theta_1(\Delta v) = \left\{ \frac{c}{2K f_0 T_r} \leq \Delta v \leq \frac{c}{f_{M-1} T_r} \right\} \tag{39}$$

When Δv is within the first-order sidelobe region Θ_1 , the maximum phase change of $\text{sinc3}(\phi_{c,i}, v; \phi_{c,i})$ can be expressed as

$$\Delta P_{max} \leq e^{j2\pi \frac{(M-1)(K-1)\Delta f}{f_{M-1}}} \tag{40}$$

Considering $\Delta f \ll f_0$, $\text{sinc3}(\phi_{c,i}, v; \phi_{c,i})$ in (34) is mainly determined by the amplitude of $\text{sinc2}(f_m, \Delta v)$.

$$\Gamma_{\text{sinc3}}(\Delta v) = N^2 \left(\sum_{m=0}^{M-1} \text{sinc2}(f_m, \Delta v) \right)^2 \tag{41}$$

Furthermore, the segmented region of $\Gamma_{\text{sinc3}}(v)$ satisfies the following characteristics.

$$\begin{aligned}
&\Gamma_{\text{sinc3}}(\Delta v_1) > \Gamma_{\text{sinc3}}(\Delta v_2), \\
&\forall \Delta v_1 \in \Theta_2(\Delta v), \forall \Delta v_2 \in \Theta_3(\Delta v)
\end{aligned} \tag{42a}$$

$$\Gamma_{\text{sinc3}}\left(\frac{c}{2T_r f_{M-1}}\right) \geq \Gamma_{\text{sinc3}}(\Delta v), \forall \Delta v \in \Theta_4(\Delta v) \tag{42b}$$

$$\Gamma_{\text{sinc3}}\left(\frac{c}{2T_r f_0}\right) \geq \Gamma_{\text{sinc3}}(\Delta v), \forall \Delta v \in \Theta_5(\Delta v) \tag{42c}$$

where,

$$\Theta_2(\Delta v) = \left\{ \frac{c}{2f_{M-1} T_r} \leq \Delta v \leq \frac{c}{2f_0 T_r} \right\} \tag{43a}$$

$$\begin{aligned}
\Theta_3(\Delta v) &= \left\{ \frac{c}{2K f_0 T_r} \leq \Delta v \leq \frac{(K-1)c}{2K f_0 T_r} \right\} \\
&\cup \left\{ \frac{(K+1)c}{2K f_{M-1} T_r} \leq \Delta v \leq \frac{c}{f_{M-1} T_r} \right\}
\end{aligned} \tag{43b}$$

$$\Theta_4(\Delta v) = \left\{ \frac{(K-1)c}{2K f_0 T_r} \leq \Delta v \leq \frac{c}{2f_{M-1} T_r} \right\} \tag{43c}$$

$$\Theta_5(\Delta v) = \left\{ \frac{c}{2f_0 T_r} \leq \Delta v \leq \frac{(K+1)c}{2K f_{M-1} T_r} \right\} \tag{43d}$$

The relationship in (42a) is established in the difference between grating and non-grating region of

$\text{sinc2}(f_m, \Delta v)$. The relationship in (42b) and (42c) is based on the monotonicity of $\text{sinc2}(f_m, \Delta v)$, $\forall m = \{0, 1, \dots, M-1\}$ within the region Θ_4 and Θ_5 . Therefore, the first-order sidelobe position satisfies the following relation.

$$\Gamma_{\text{sinc3}}(\Delta v_{1\text{-side}}) \geq \Gamma_{\text{sinc3}}(\Delta v), \quad \exists \Delta v_{1\text{-side}} \in \Theta_2(\Delta v), \forall \Delta v \in \Theta_1(\Delta v) \quad (44)$$

When $f_0 \gg \Delta f$, Θ_2 can be regarded as a neighborhood of $\Delta v_{1\text{-side}}$. Since $\Gamma_{\text{sinc3}}(\Delta v)$ is a continuous function, the neighborhood Θ_2 could describe the local characteristics of $\Gamma_{\text{sinc3}}(\Delta v_{1\text{-side}})$. In that case, $\Gamma_{\text{sinc3}}(\Delta v)$ satisfies as follows:

$$\Gamma_{\text{sinc3}}(\Delta v) \approx \Gamma_{\text{sinc3}}(\Delta v_{1\text{-side}}), \forall \Delta v \in \Theta_2(\Delta v) \quad (45)$$

To simplify the subsequent processing, $\Delta v = \frac{c}{2(f_0 + (M-1)/2\Delta f)T_r} \in \Theta_2(\Delta v)$ as the position of the first-order sidelobe. Substituting $\Delta v = \frac{c}{2(f_0 + (M-1)/2\Delta f)T_r}$ into (32), the first-order sidelobe of $\text{sinc3}(\phi_{c,i}, v; \phi_{c,i})$ can be expressed as

$$\begin{aligned} & \text{sinc3}(\phi_{c,i}, v_a \cos(\phi_{c,i}) - \frac{c}{2(f_0 + (M-1)/2\Delta f)T_r}; \phi_{c,i}) \\ &= N^2 \left(\sum_{k=0}^{K-1} \frac{\sin(M\pi \frac{\Delta f}{f_0 + \frac{M}{2}\Delta f} k)}{\sin(\pi \frac{\Delta f}{f_0 + \frac{M}{2}\Delta f} k)} \right)^2 \\ &\approx N^2 M^2 \gamma(\Delta f; f_0, M, K)^2 \end{aligned} \quad (46)$$

where,

$$\gamma(\Delta f; f_0, M, K) \triangleq \sum_{k=0}^{K-1} \text{sinc} \left(\frac{kM\Delta f}{f_0 + \frac{M}{2}\Delta f} \right) \quad (47)$$

Under the condition $\Delta f \ll f_0$, the approximate relationship in (46) is satisfied. Furthermore, substitute (46) into (29) and the power of the first-order sidelobe of $P(\phi_{c,i}, v)$ can be represented as

$$P_{1\text{-side}} = \frac{\sigma_n^2(\sigma_n^2 + MNK\sigma_{c,l}^2)}{MNK\sigma_n^2 + M^2K^2N^2\sigma_{c,l}^2 - \sigma_{c,l}^2N^2M^2\gamma(\Delta f)^2} \quad (48)$$

When $\sigma_{c,i}^2$ is much greater than σ_n^2 , (48) can be approximated as follows:

$$P_{1\text{-side}} \approx \frac{\sigma_n^2}{MNK(1 - \gamma(\Delta f)^2/K^2)} \quad (49)$$

The $P_{1\text{-side}}$ in (49) indicates that the clutter power does not affect the power spectrum $P(\phi, v)$ in the velocity ambiguity period, which is only related to the noise power and the configuration of the FDA-MIMO radar. When $\Delta v = 0$, the power of mainlobe in $P(\phi, v)$ is expressed as

$$P_{\text{main}} = \frac{\sigma_n^2}{MNK} + \sigma_c^2 \quad (50)$$

Since the power of the first-order sidelobe represents the power of ambiguity clutter, peak-to-sidelobe-rate (PSLR) is defined to measure the clutter suppression

performance of FDA-MIMO.

$$\begin{aligned} \chi_{fda} &= \frac{P_{\text{main}}}{P_{1\text{-side}}} \\ &= \frac{\frac{\sigma_n^2}{MNK} + \sigma_c^2}{\frac{\sigma_n^2(\sigma_n^2 + MNK\sigma_c^2)}{MNK\sigma_n^2 + M^2K^2N^2\sigma_c^2 - \sigma_c^2N^2M^2\gamma(\Delta f)^2}} \end{aligned} \quad (51)$$

where χ_{fda} denotes the lower bound of ambiguity clutter suppressed by FDA-MIMO radar. Through simplifying (51), we can get the following result.

$$\Delta f = \gamma^{-1} \left(\sqrt{K^2 - \frac{K\sigma_n^2(\chi_{fda} - 1)}{MN\sigma_c^2}} \right) \quad (52)$$

where, $\gamma^{-1}(\cdot)$ represents the inverse function of $\gamma(\cdot)$. The formula in (52) describes the relationship between ambiguity clutter and frequency increment of FDA-MIMO-STAP radar.

IV. The Strategy of Resolving Ambiguity

In this section, the method for Doppler ambiguity of the high-speed target using the FDA-MIMO radar is analyzed. We consider the scenario and target signal model in Section II.A. Furthermore, the high-speed moving target signal received by FDA-MIMO radar can be expressed as

$$\mathbf{y} = \bar{\mathbf{y}}_s + \mathbf{y}_n \quad (53)$$

In (53), $\bar{\mathbf{y}}_s$ represents the FDA-MIMO target signal compensated by target range information and it can be derived as

$$\begin{aligned} \bar{\mathbf{y}}_s &= \xi_s \mathbf{h}(r_s) \odot \text{vec} \{ \mathbf{A}_{TRD}(\phi_s, r_s, \\ & \quad v_a \cos(\phi_s) - v_s \cos(\Psi_s) \} \gamma_s[l; r_s]; \end{aligned} \quad (54)$$

Through a similar derivation from (21) to (29), the power spectrum of FDA-MIMO target signal can be expressed as follows:

$$P_s(\phi, v) = \frac{\sigma_n^2(\sigma_n^2 + MNK\sigma_{s,L}^2)}{MNK\sigma_n^2 + M^2K^2N^2\sigma_{s,L}^2 - \sigma_c^2 \text{sinc4}(\phi, v)} \quad (55)$$

where,

$$\sigma_{s,L}^2 = \sigma_s^2 / L \quad (56a)$$

$$\text{sinc4}(\phi, v) = \text{sinc3}(\phi, v + v_s \cos(\Psi_s); \phi_s) \quad (56b)$$

To ensure that the target can be detected after the ambiguity clutter suppression, the power of the target must be higher than that of the ambiguity clutter. In that case, the mainlobe power of target need to be greater than the first-order sidelobe of clutter.

$$\frac{\sigma_n^2}{MNK} + \sigma_{s,L}^2 \geq P_{1\text{-side}} \quad (57)$$

Considering $\sigma_{s,L}^2$ is strongly correlated to the number of sample points L , an approach is to modify L in order to meet the requirements in (57). Therefore, a sampling function as follows is used to reduce the total number of sampling points L .

$$\mathbf{D}_s = [\mathbf{0}_{L_0 \times l_0}, \mathbf{I}_{L_0 \times L_0}, \mathbf{0}_{L_0 \times (L - L_0 - l_0)}]^H \quad (58)$$

where l_0 satisfies $l_0 < l_{r,s} < l_0 + L_0$. Further, perform downsampling processing on $\bar{\mathbf{y}}_s$ using $\mathbf{D}_s$ as the reference.

$$\hat{\mathbf{y}}_s = \bar{\mathbf{y}}_s \mathbf{D}_s \quad (59)$$

Substitute (59) to (53) and perform subsequent actions. Then the power spectrum in (55) can be rewritten as

$$P_s(\phi, v) = \frac{\sigma_n^2(\sigma_n^2 + MNK\sigma_{s,L_0}^2)}{MNK\sigma_n^2 + M^2K^2N^2\sigma_{s,L_0}^2 - \sigma_c^2 \text{sinc}^4(\phi, v)} \quad (60)$$

To describe the ability of resolving Doppler ambiguity for FDA-MIMO radar, the maximum non-ambiguity range in velocity dimension is need to be determined. It is worth noting that the maximum non-ambiguity range of $P_s(\phi, v)$ is equal to the minimum blurred period of $\text{sinc}^4(\phi, v)$. Meanwhile, the velocity dimension period of sinc^3 and sinc^4 are all the same. Therefore, the maximum non-ambiguity range of $P_s(\phi, v)$ in velocity dimension satisfies the following condition.

$$\begin{aligned} v_{max} &= IT_v(\text{sinc}^2(f_m, \Delta v)), \\ \exists I \in \mathbb{Z}^+, \forall m \in \{0, \dots, M-1\} \end{aligned} \quad (61)$$

where v_{max} denotes the maximum non-ambiguity range of $P_s(\phi, v)$ in velocity dimension and $T_v(\cdot)$ represents the velocity period of inside element. Considering that there is the close-form in $T_v(\text{sinc}^2(f_m, \Delta v))$, the following relationship between v_{max} and parameters of FDA-MIMO radar is satisfied.

$$\begin{aligned} I &= v_{max} / \left(\frac{c}{2f_m T_r} \right) - v_{max} / \left(\frac{c}{2f_{m-1} T_r} \right) \\ &= v_{max} / \left(\frac{c}{2\Delta f T_r} \right), \exists I \in \mathbb{Z}^+ \end{aligned} \quad (62)$$

According to characteristics of positive integer I , it can be known that there is a lower bound in v_{max} .

$$v_{max} = I \left(\frac{c}{2\Delta f T_r} \right) \geq \left(\frac{c}{2\Delta f T_r} \right), I \in \mathbb{Z}^+ \quad (63)$$

When f_0 is much greater than Δf , $\frac{c}{2\Delta f T_r}$ is also greater than $\frac{c}{2f_0 T_r}$ which means that the maximum non-ambiguity range of $P_s(\phi, v)$ in velocity dimension is drastically expanded. At this time, the velocity ambiguity of high-speed moving target can be effectively resolved.

V. NUMERICAL SIMULATIONS

In this section, numerical simulations are conducted to verify the correctness of the derived characteristics of Doppler-ambiguous clutter in FDA-MIMO radar and the effectiveness of the proposed strategy for resolving target Doppler ambiguity. In these experiments, the movement speed of aircraft v_a is set to 300 m/s and the clutter distribution as shown in Fig.1. The default simulation parameters of the FDA-MIMO radar system are listed in TABLE I.

TABLE I: DEFAULT SIMULATION PARAMETERS

Parameter	Symbol	Value
Carrier frequency	f_0	3GHz
Wavelength	λ_0	0.1m
Transmit element number	M	8
Receive element number	N	8
Pulses number	K	8
Element spacing	d	0.05m
Frequency increment	Δf	10MHz
Pulse repetition time	T_r	50us

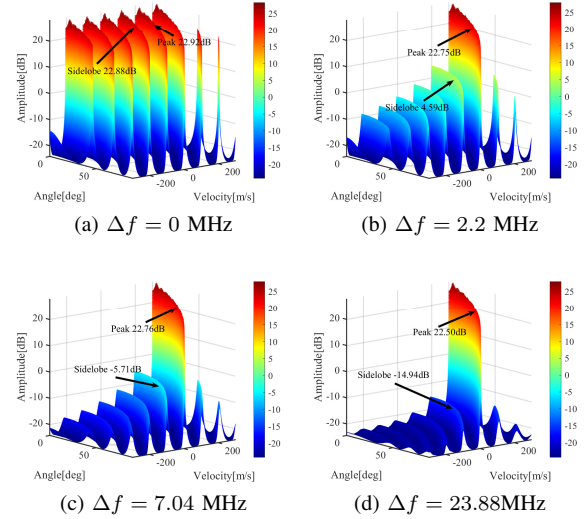


Fig. 2: Clutter power spectrum of FDA-MIMO radar with different frequency increment.

A. The PSLR of Clutter Power Spectrum of FDA-MIMO Radar

In this subsection, we verify the validity of the relationship between ambiguity clutter PSLR and frequency increment derived in (52) by presenting four sets of normalized clutter power spectrum at different frequency increments. We set the clutter-to-noise ratio (CNR) to 20 dB. It can be seen from Fig.2, under airborne side-looking conditions, the clutter ridge exhibits an arc-shaped profile which means its velocity varies as a cosine function of LOS angle shown in (13). Meanwhile, in addition to the true clutter ridge, there is periodical ambiguous clutter ridges in the velocity dimension. When Δf is set to 0, the clutter ridges corresponding to different ambiguous periods show no obvious difference. The mainlobe and sidelobe of the clutter power spectrum are nearly identical, resulting in the PSLR close to 0 dB. As Δf increases, the power of the ambiguous clutter ridges gradually decreases, which leads to a gradual rise in the PSLR of the clutter power spectrum. When Δf increases to 2.02 MHz, 7.04 MHz, and 23.8 MHz, respectively, PSLRs of the clutter power spectrum become 18.16 dB, 28.47 dB,

and 37.44 dB. In the meantime, PSLRs calculated with (52) are 20 dB, 30 dB, and 40 dB, respectively, which are only slightly higher than the measured results. This may be attributed to the sidelobe superposition of multiple clutter scattering points. From the above results, it can be seen that the closed-form relationship between PSLR and the frequency increment derived in this paper can characterize the PSLR behavior of the FDA-MIMO clutter spectrum efficiently.

B. The Performance of FDA-MIMO Ambiguity Clutter Suppression

In this section, the influencing factors of the suppression performance of FDA-MIMO radar ambiguity clutter are investigated. First, we compare the performance of Doppler-ambiguous clutter suppression among different radar. The signal-to-clutter-plus-noise ratio (SCNR) loss is chosen as the metric to evaluate the ambiguous clutter suppression performance. A higher SCNR loss indicates better clutter suppression performance. The specific definition of SCNR loss is given as follows:

$$\text{SCNR}_{\text{loss}} = \frac{\text{SCNR}_{\text{out}}}{\text{SNR}_{\text{ideal}}} \quad (64)$$

where $\text{SCNR}_{\text{loss}}$ denotes SCNR loss, SCNR_{out} represents the output SCNR after filtering by STAP filter and $\text{SNR}_{\text{ideal}}$ is the ideal SNR in the absence of clutter. Then, the relationship between the SCNR loss of the FDA-MIMO radar and the frequency offset and CNR is also investigated. The target velocity and LOS angle are set to 137.43 m/s and 50° , respectively. For convenience in the following description, we define $V_1(v)$ as the velocity interval corresponding to the true clutter ridge and $V_2(v)$ as the velocity interval corresponding to the ambiguity clutter ridge.

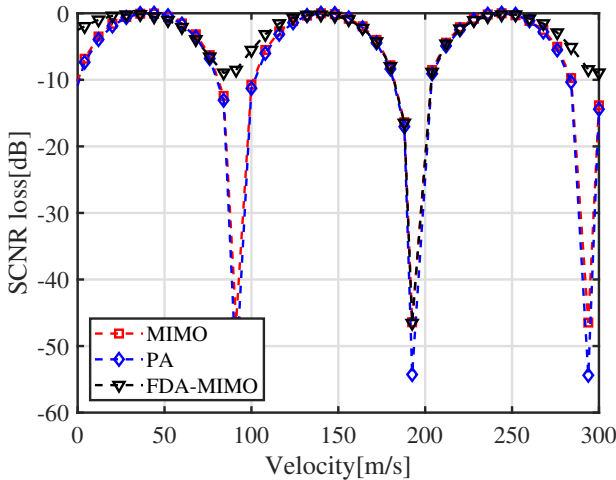


Fig. 3: The SCNR loss curve comparison for different radars.

Fig. 3 shows the SCNR loss curves for different radars. It can be seen that in the $V_1(v)$ region, the SCNR loss

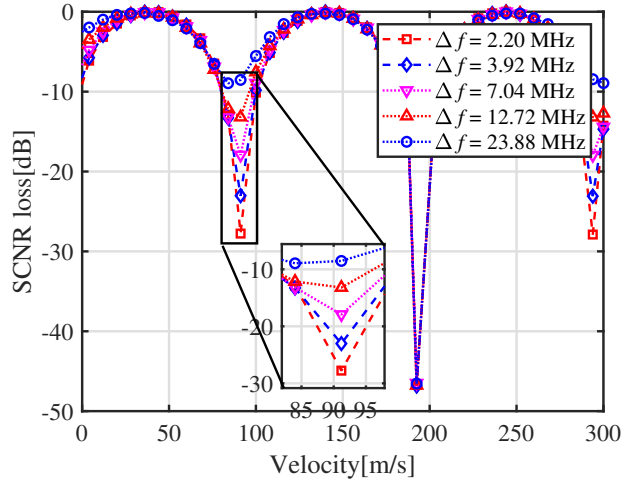


Fig. 4: The SCNR loss of FDA-MIMO radar with different Δf .

of the three radars are relatively close. Only the PA radar exhibits a lower SCNR loss than the other two radars. However, in the $V_2(v)$ region, the SCNR loss of the FDA-MIMO radar is much higher than that of the PA radar and the conventional MIMO radar. This indicates that the Doppler-ambiguous clutter suppression capability of the FDA-MIMO radar is much higher than that of the PA radar and the conventional MIMO radar.

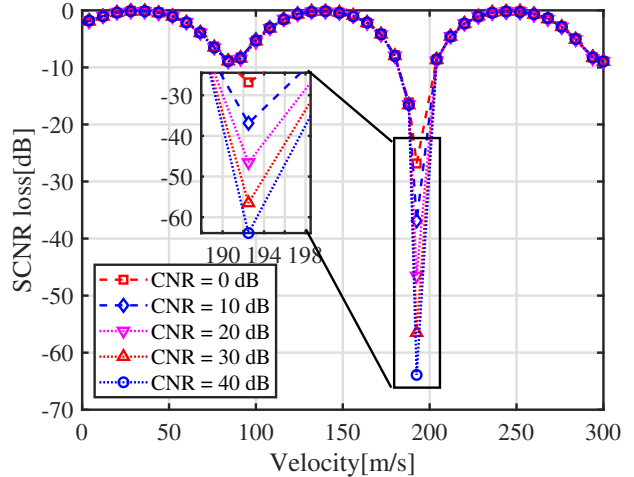


Fig. 5: The SCNR loss of FDA-MIMO radar with different CNR.

Fig. 4 shows the SCNR loss curves of FDA-MIMO radar with different frequency increment. In this experiment, we set CNR to 20 dB and only investigate the influence of frequency increment on the ambiguity clutter suppression performance. We compare the SCNR loss curves of the FDA-MIMO radar with frequency increment of 2.20, 3.92, 7.04, 12.72, and 23.88 MHz, which correspond to the FDA-MIMO radar clutter with theoretical PSLRs of 20, 25, 30, 35, and 40 dB, re-

spectively. When the frequency offset differs, the SCNR loss in the $V_1(v)$ region remains essentially unchanged. The suppression capability of the FDA-MIMO radar for real clutter is basically consistent. However, as the frequency increment increases, the SCNR loss in $V_2(v)$ region gradually increases, which means the capability of the FDA-MIMO radar to suppress ambiguous clutter in velocity dimension is gradually enhanced. Meanwhile, as the frequency increment increases, the SCNR loss in the $V_2(v)$ region also increases essentially in increments of 5 dB, which is consistent with the result derived in (52). The above results verify that the capability of FDA-MIMO radar to suppress ambiguity clutter in velocity dimension. Furthermore, it also demonstrates that the larger frequency increment leads to better ambiguity clutter suppression performance.

To investigate the influence of CNR on the Doppler-ambiguous clutter suppression performance, we conduct multiple experiments with the same frequency increment 23.88MHz but different CNRs. The corresponding results are shown in Fig.5. It can be seen that SCNR loss in $V_2(v)$ region remain the high level which means the high output SCNR filtering by STAP filter. It indicates that the ambiguity clutter is suppressed to a certain extent when $\Delta f \neq 0$. When the CNR varies, the SCNR loss in $V_1(v)$ changes significantly after STAP filtering and the higher CNR causes the deeper notches of SCNR loss curve. However, the SCNR loss in $V_2(v)$ region only changes slightly. It indicates that the Doppler-ambiguous clutter suppression capability of the FDA-MIMO radar does not fluctuate significantly with the change of CNR, which is consistent with the conclusion of (49). This illustrates that the ambiguity clutter suppression performance in velocity dimension is insensitive to the variation of CNR for FDA-MIMO radar.

C. The Verification of FDA-MIMO Resolving Doppler Ambiguity

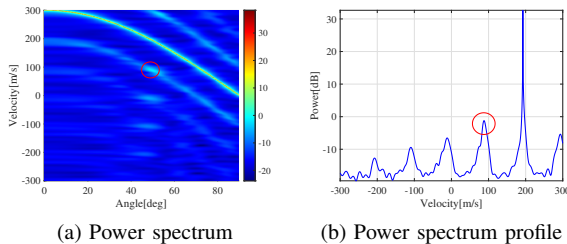


Fig. 6: Result of Doppler ambiguity resolution for the FDA-MIMO radar received signal.

In this subsection, we investigate the Doppler ambiguity resolution of high-speed targets for FDA-MIMO radar. The azimuth and elevation angles of the real target are set to $\theta_s = 50^\circ$ and $\varphi_s = 6.8^\circ$, respectively, with the velocity of 91.3 m/s along the direction of the airborne platform motion. Correspondingly, the real LOS angle ϕ_s and

velocity of the target in the airborne radar power spectrum are 47° and 192 m/s, respectively. At the same time, CNR and SNR are set to 20dB and 0dB, respectively.

Fig.6 shows the received signal power spectrum when the frequency increment is set to 23.88 MHz. It can be observed that, in addition to the true clutter ridge, there exists another distinguishable point in the power spectrum (marked by the red circle). By further selecting the LOS angle of 50° and extracting the velocity dimension from the power spectrum in Fig. 4a, the result in Fig. 4b is obtained. It can be seen that the power corresponding to the actual target velocity is only lower than the actual clutter power, and higher than the power of other ambiguous clutter, enabling target detection in the presence of ambiguous clutter. Meanwhile, the target LOS angle ϕ_s and velocity v_s obtained from the power spectrum are 47° and 192.5 m/s, respectively, which achieves accurate estimation of the true velocity of a high-speed target under the velocity ambiguity period of 100 m/s. Therefore, the above results validate the effectiveness of the proposed Doppler ambiguity resolution method for high-speed targets using FDA-MIMO radar.

D. The Performance of Target Detection in Doppler-ambiguous Clutter

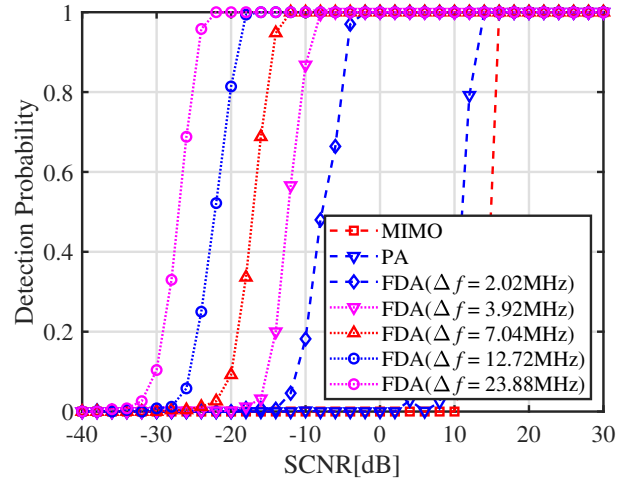


Fig. 7: The detection probability of FDA-MIMO radar with different Δf .

In this section, we conduct multiple sets of Monte Carlo experiments to measure the target detection probability under different radars, different frequency increments, and SCNR conditions, aiming to investigate the influence on the target detection performance of FDA-MIMO radar in a clutter environment. In each Monte Carlo experiment, 500 trials are performed. The target position and motion parameters are exactly the same as those set in Section V.C.

The experimental results are shown in Fig.7. It can be observed that for the FDA-MIMO radar at any frequency

increment, the target detection probability gradually increases with the SCNR and is higher than that of the PA radar and the conventional MIMO radar. As the frequency increment increases, the SCNR required to achieve the same detection probability gradually decreases. This indicates that a larger frequency increment provides the FDA-MIMO radar with a stronger capability to suppress ambiguous clutter. In this case, even if the clutter power becomes stronger, the FDA-MIMO radar can sufficiently reduce the influence of Doppler-ambiguous clutter. Therefore, under a low SCNR, the target detection probability can reach the same level as that under a high SCNR when a lower frequency increment is used. Meanwhile, these results further validate the conclusion drawn in Section V.B that the increase in frequency increment enhances the capability of the FDA-MIMO radar to suppress ambiguous clutter in the velocity dimension. It is worth noting that, for FDA-MIMO radars with different frequency increments, the SCNR differences under the same detection probability are all approximately 5 dB. It is consistent with the SCNR performance under different frequency offsets presented in Section V.B. Based on the above results, it can be concluded that the target detection performance of the FDA-MIMO radar is superior to that of the PA radar and the conventional MIMO radar in the clutter environment with the same SCNR. Meanwhile, an increase in the frequency increment enhances the target detection probability of the FDA-MIMO radar in the Doppler-ambiguous clutter environment and the specific amount of enhancement can be quantified by the theoretical lower bound of the Doppler-ambiguous clutter suppression performance derived in (52).

VI. CONCLUSION

In this paper, we have proposed a strategy for resolving Doppler ambiguity of high-speed targets using FDA-MIMO-STAP radar. With this strategy, the FDA-MIMO radar is capable of performing the dual tasks of Doppler-ambiguous clutter suppression and Doppler ambiguity resolution for high-speed moving targets. We have also derived the closed-form relationship of PSLR of the clutter power spectrum for the FDA-MIMO radar. Based on this characteristic, we further determined the theoretical lower bound of the Doppler-ambiguous clutter suppression performance for the FDA-MIMO-STAP radar. Finally, we derived the theoretical lower bound for resolving Doppler ambiguity of high-speed targets with the FDA-MIMO radar. Experimental results validate the effectiveness of the proposed strategy and the correctness of the derived performance bounds. Future work could be devoted to the derivation of more accurate performance bounds for FDA-MIMO radar and validation with measured data.

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REFERENCES

- [1] P. Klemm
Applications of Space-Time Adaptive Processing. London, U.K.: IET, 2004.
- [2] P. Klemm
Principles of Space-Time Adaptive Processing. London, U.K.: IET, 2006.
- [3] L. Brennan and L. Reed
Theory of adaptive radar
IEEE Transactions on Aerospace and Electronic Systems, vol. AES-9, no. 2, pp. 237–252, 1973.
- [4] J. Ward
Space-time adaptive processing for airborne radar
In *IEE Colloquium on Space-Time Adaptive Processing (Ref. No. 1998/241)*, 1998, pp. 2/1–2/6.
- [5] Y. Wang, K. Duan, and W. Xie
Cross beam stap for nonstationary clutter suppression in airborne radar
International Journal of Antennas and Propagation, vol. 2013, pp. 112–124, 01 2013.
- [6] W. Xie, K. Duan, F. Gao, Y. Wang, and Z. Zhang
Clutter suppression for airborne phased radar with conformal arrays by least squares estimation
Signal Processing, vol. 91, no. 7, pp. 1665–1669, 2011.
- [7] R. Raghavan, H. Qiu, and D. McLaughlin
Cfar detection in clutter with unknown correlation properties
IEEE Transactions on Aerospace and Electronic Systems, vol. 31, no. 2, pp. 647–657, 1995.
- [8] J. Xu, S. Zhu, and G. Liao
Range ambiguous clutter suppression for airborne fda-stap radar
IEEE Journal of Selected Topics in Signal Processing, vol. 9, no. 8, pp. 1620–1631, 2015.
- [9] Y. Wang, S. Zhu, G. Liao, L. Lan, Z. Chen, and F. Liu
Resolving doppler ambiguity via spread phase alignment in fda-mimo radar
In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.
- [10] W. Chen, W. Xie, and Y. Wang
Range-dependent ambiguous clutter suppression for airborne ssf-stap radar
IEEE Transactions on Aerospace and Electronic Systems, vol. 58, no. 2, pp. 855–867, 2022.
- [11] Y. Wang, L. Zhou, J. Wen, J. Yuan, and Z. Huang
Range ambiguity clutter suppression method for space-air based bistatic radar
In *2018 3rd International Conference on Mechanical, Control and Computer Engineering (ICMCCE)*, 2018, pp. 355–360.
- [12] X. He, G. Liao, S. Zhu, J. Xu, and J. Zhu

- Range ambiguous clutter suppression approach with elevation time diverse array
IEEE Transactions on Aerospace and Electronic Systems, vol. 58, no. 1, pp. 359–373, 2022.
- [13] Z. Qiu, J. He, and K. Duan
Range ambiguous clutter suppression for airborne radar using pulse phase coding
In *2024 IEEE Radar Conference (RadarConf24)*, 2024, pp. 1–6.
- [14] Y. Wang, S. Zhu, X. He, X. Li, X. Wu, and Y. Guo
A range-ambiguous clutter suppression method via epc-mimo bistatic airborne radar
In *2022 5th International Conference on Information Communication and Signal Processing (ICICSP)*, 2022, pp. 344–348.
- [15] Z. Qiu, K. Duan, and Z. Liao
Range ambiguous clutter suppression for forward-looking airborne radar based on frequency coding and phase waveforms modulation in slow-time
IEEE Transactions on Aerospace and Electronic Systems, vol. 60, no. 6, pp. 8470–8482, 2024.
- [16] P. Lu and Z. Wang
Range-ambiguous clutter suppression by phase-coding and subspace projection
In *2025 5th International Conference on Sensors and Information Technology*, 2025, pp. 164–168.
- [17] P. Antonik, M. C. Wicks, H. D. Griffiths, and C. J. Baker
Frequency diverse array radars
In *Proc. IEEE Radar Conf.*, 2006, pp. 215–217.
- [18] Y. B. Wang, W. Q. Wang, and H. Z. Shao
Frequency diverse array radar cramér-rao lower bounds for estimating direction, range, and velocity
Int. J. Antenn. Propag., vol. 2014, pp. 1–15, Apr. 2014.
- [19] C. Cui, J. Xiong, W. Q. Wang, and W. Wu
Localization performance analysis of FDA radar receiver with two-stage estimator
IEEE Trans. Aerosp. Electron. Syst., vol. 54, no. 6, pp. 2873–2887, Dec. 2018.
- [20] W. Q. Wang and H. C. So
Transmit subaperturing for range and angle estimation in frequency diverse array radar
IEEE Trans. Signal Process., vol. 62, no. 8, pp. 2000–2011, Apr. 2014.
- [21] J. W. Xu, G. S. Liao, S. Q. Zhu, L. Huang, and H. C. So
Joint range and angle estimation using MIMO radar with frequency diverse array
IEEE Trans. Signal Process., vol. 63, no. 13, pp. 3396–3410, Jul. 2015.
- [22] Y. Z. Wang and S. Q. Zhu
Main-beam range deceptive jamming suppression with simulated annealing FDA-MIMO radar
IEEE Sensors J., vol. 20, no. 16, pp. 9056–9070, Aug. 2020.
- [23] Y. B. Liu, C. Y. Wang, J. Gong, and G. Chen
Discrimination of mainlobe deceptive target with meter-wave FDA-MIMO radar
IEEE Commun. Lett., vol. 26, no. 5, pp. 1131–1135, May. 2022.
- [24] Y. Wang and S. Zhu
Range ambiguous clutter suppression for fda-mimo forward looking airborne radar based on main lobe correction
IEEE Transactions on Vehicular Technology, vol. 70, no. 3, pp. 2032–2046, 2021.
- [25] D. Song, S. Chen, Q. Feng, F. Xi, and Z. Liu
Subspace-based stap in airborne fda radar for homogeneous clutter under range ambiguity
IEEE Transactions on Aerospace and Electronic Systems, vol. 60, no. 3, pp. 2909–2921, 2024.
- [26] Y. Sun, S. Shao, W.-Q. Wang, M. S. Greco, F. Gini, and S. Zhang
Nonlinear frequency offset optimization strategy for solving secondary range ambiguity in planar fda-stap radar
IEEE Transactions on Aerospace and Electronic Systems, vol. 61, no. 3, pp. 8054–8069, 2025.
- [27] Y. Wu, B. Jiu, Y. Zhang, H. Zheng, B. Yang, and H. Liu
An fda-mimo-based range-ambiguous clutter sensing approach for stap
In *2023 IEEE International Radar Conference (RADAR)*, 2023, pp. 1–6.
- [28] D. Song, S. Chen, Q. Feng, F. Xi, and Z. Liu
Subspace-based stap in airborne fda radar for homogeneous clutter under range ambiguity
IEEE Transactions on Aerospace and Electronic Systems, vol. 60, no. 3, pp. 2909–2921, 2024.
- [29] R. Gui, W.-Q. Wang, A. Farina, and H. C. So
Fda radar with doppler-spreading consideration: Mainlobe clutter suppression for blind-doppler target detection
Signal Processing, vol. 179, p. 107773, 2021.